

Question Paper

Exam Date & Time: 21-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Social Networks [CSE_4204]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate how **adjacency matrices** and **edge lists** represent the same network structure, and **compare** their effectiveness for various types of **network analysis** tasks. **Assess** the advantages and limitations of each representation method, and **justify** which format provides better insights for analyzing complex or large-scale networks using a suitable example. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply appropriate **data preprocessing techniques** to **illustrate** how missing or noisy data in real-world network datasets can affect the accuracy of **network analysis** and **community detection** results. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Apply **Gephi** or **NetworkX** to **construct** and **visualize** a **social or communication network**, and **identify** key structural properties such as **node degree**, **clustering**, and **network density** to show how relationships among individuals are represented. (2)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Evaluate the **strength of weak ties** in facilitating **information flow** and **opportunity discovery** within social structures. **Assess** how weak connections contribute to bridging otherwise disconnected communities and justify their role in enhancing network diversity. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply the concept of **homophily** to **identify** how similarities among individuals-such as shared interests or demographics-**influence** the formation of social connections within a network. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Apply the concept of **balance theory** to **identify** how positive and negative relationships influence the **stability** of small social groups. (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Evaluate the **effectiveness** of different **centrality measures**-degree, closeness, betweenness, and eigenvector-in identifying **influential nodes** within a network. **Justify** which measure provides the most comprehensive insight for real-world applications like social influence or information diffusion. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Apply the concept of **hubs and authorities** to **identify** how information importance is distributed in a web or citation network, and **illustrate** how the **HITS algorithm** helps distinguish between content creators (hubs) and trusted sources (authorities). (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Apply the concept of **cascading behavior** to **identify** how an initial piece of information can trigger a chain reaction of adoption or sharing within a social network. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Compare and **justify** how the **rich-get-richer** mechanism leads to the formation of **scale-free structures** in growing networks. **Assess** its implications for **inequality** and **node influence** within large-scale information or social systems. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Construct examples of **different network topologies**-such as random, small-world, and scale-free networks-to **demonstrate** how structural differences influence the **speed and reach** of epidemic spreading. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Construct a simple example to **illustrate** how the **power-law degree distribution** characterizes node connectivity in a **scale-free network**. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Evaluate how the **small-world property** influences **information diffusion, collaboration, and innovation** in large social or technological networks. **Propose** how understanding this property can be applied to **optimize communication systems** or **design resilient network structures**. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply your understanding of **information diffusion models** to **construct** a set of **network-based strategies**-such as targeting pseudocores or key connectors-that can **enhance the likelihood of viral spread** in digital marketing or awareness campaigns. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Apply your understanding of **viral diffusion** to **interpret** one real-world case-such as the spread of a popular hashtag or challenge-and explain how **network structure** contributed to its rapid growth. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_4204:Social Networks

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4204:Social Networks

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	9	22(44%)
4	Analysis	6	28(56%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)